Integration Flow – Summary of Work

Module Context

This integration is developed as part of the **Project Portfolio Management (PPM)** process in Oracle Fusion Applications. It enables the **automated import of project-related Award Compensations**, approved through the workflows, and ensures those costs are correctly transferred to the **Payroll module** for processing.

---

1. Integration Development

An **App-Driven Orchestration Integration** was implemented using **Oracle Integration Cloud (OIC)**. The integration automates the flow of Approved Project Award Compensation expenses from the PPM module to the Payroll module. It includes the following components:

- **BI Report Invocation**
  The integration calls a **BI Report** developed in Oracle Fusion to retrieve all **approved** Project Award Compensation.
  The report is employee-specific and project-specific, ensuring a detailed breakdown of expenses for each project under PPM.
- **FBDI File Generation**
  The fetched data is transformed into an **FBDI (File-Based Data Import)** file, adhering to the template structure of the **Import Costs** ESS job.
  This ensures compatibility with the Oracle Fusion PPM and Payroll import process.
- **ESS Job Execution – Import Costs**
  The integration invokes the **Import Costs ESS job** in Oracle Fusion, which imports the Project Award Compensation costs into the relevant project records.
  These costs represent amounts borne by the company on behalf of employees, and they are attributed to the respective **projects and employees** in PPM.
  The data is also made available for downstream processing in the **Payroll module**.
- **Error Handling with Notifications**
  A comprehensive error-handling mechanism is built into the integration.
  In case of any processing failure (e.g., report fetch failure, transformation errors, or ESS job failure), an **email notification** is sent automatically to alert the responsible team or individual.

---

2. Integration Endpoint

- The **REST endpoint URL** of the integration was generated and securely saved.
  This endpoint is triggered by the custom ESS job, ensuring seamless execution without manual input.

3. ESS Job Triggering

- A **Custom ESS Job** was configured within Oracle Fusion to serve as the trigger mechanism for the integration.
  - The ESS job is scheduled or invoked manually as needed.
  - It eliminates the need for human intervention and enables **event-based or scheduled automation** of the cost import process.

Custom ESS Job Configuration

1. BI Report Setup

- A **BI Publisher Data Model** of type **HTTP XML Feed** was created.
  This data model connects directly to the integration's HTTP endpoint.
- A **BI Report** was built on top of this data model.
  This report acts as a bridge between the ESS job and the integration, allowing report-based triggering of the integration flow.

2. ESS Job Creation

- A **Custom ESS Job** was created in Oracle Fusion Applications using the path of the BI report.
  - The job is configured to call the integration automatically.
  - It facilitates the **scheduled or manual import of** Project Award Compensation **costs** into Oracle Fusion.

End Result & Business Impact

- The solution ensures that **approved** Project Award Compensation **costs associated with projects** are automatically transferred from PPM to Payroll.
- It reduces manual effort, eliminates errors, and enhances traceability of project-related expenses by **accurately attributing costs per employee and per project**.
- It ensures the organization meets its financial reporting and payroll processing requirements in a timely and automated manner

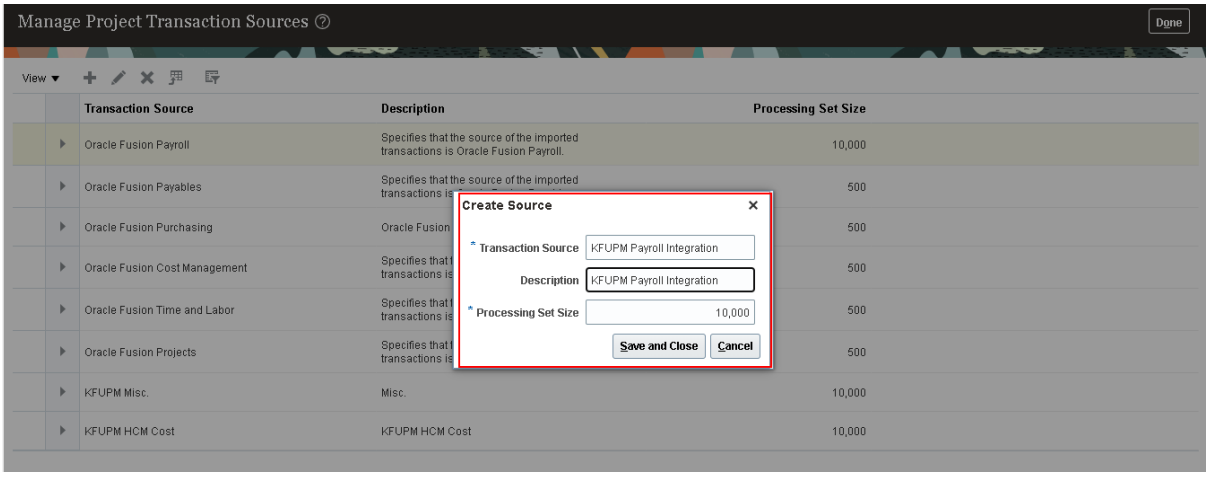
Functional Configuration:

We need to configure the **Transaction Source**, **Document**, and **Document Entry** since we are using specific source, document, and document entry for the records imported through the integration (OIC).

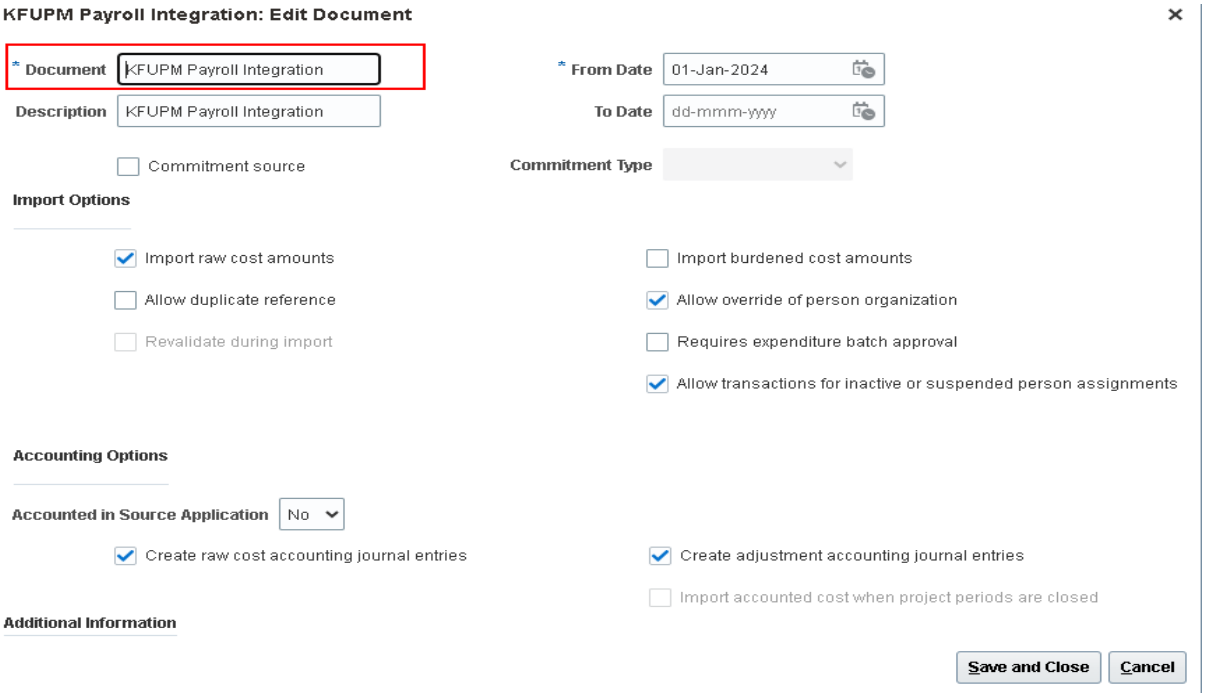
NAME: "KFUPM Payroll Integration"(Same Name For **Source**, **Document**, and **Document Entry**)

Navigation >> Setup and Maintenance >>Task >>Manage Project Transaction Sources

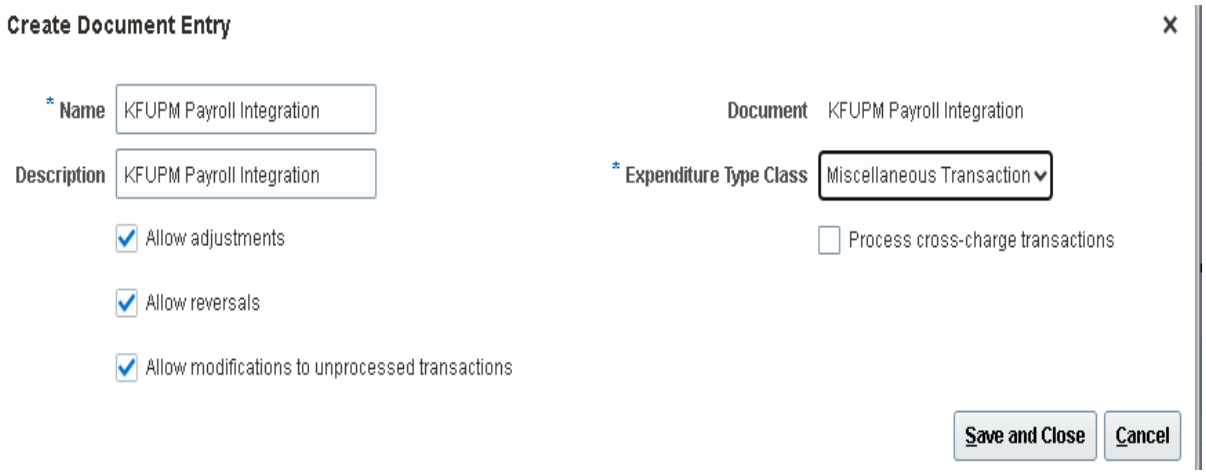
Create Source :



Create Document:



Create Document Entry :



You can view the created Transaction Source**,** Document**, and** Document Entry **in the image below.**

KFUPM Payroll Integration	KFUPM Payroll Integration	10,000

KFUPM Payroll Integration: Documents

View

Document	Description	From Date	To Date	Commitment Source	Commitment Type	Import Raw Cost Amounts	Import Burdened Cost Amounts	Allow Duplicate Reference	Allow Override of Person Organization
KFUPM Payroll Integration	KFUPM Payroll I...	01-Jan-2024				✓			✓

KFUPM Payroll Integration: KFUPM Payroll Integration: Document Entries

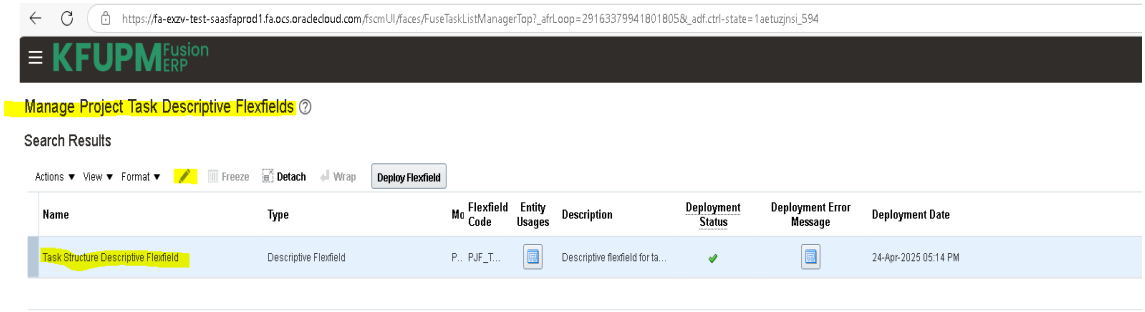
View

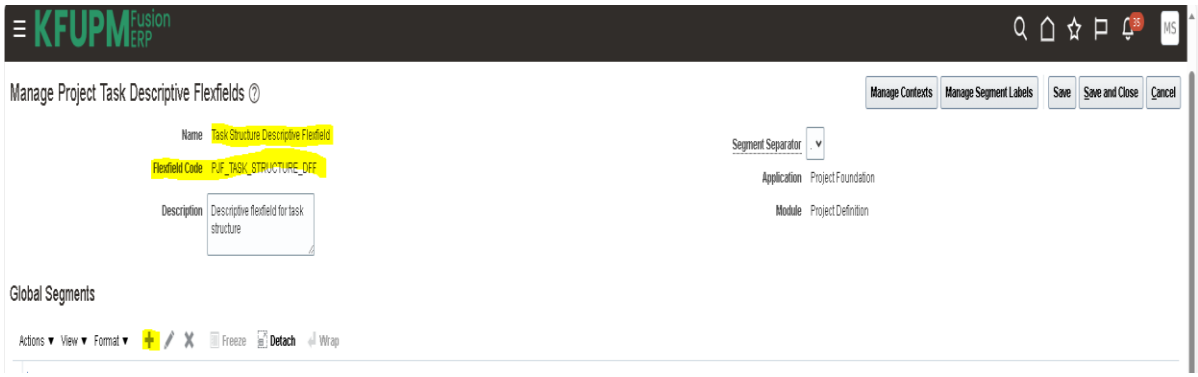
Name	Description	Allow Adjustments	Allow Reversals	Allow Modifications to Unprocessed Transactions	Process Cross-Charge Transactions	Expenditure Type Class
KFUPM Payroll Integrati...	KFUPM Payroll Integration	✓	✓	✓		Miscellaneous Transaction

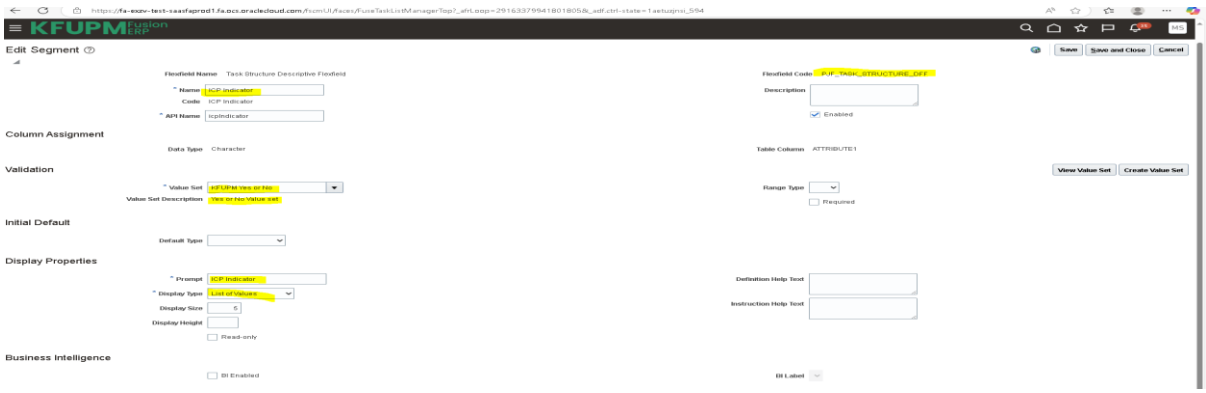
DFF Configuration (DFF Name: ICP Indicator)

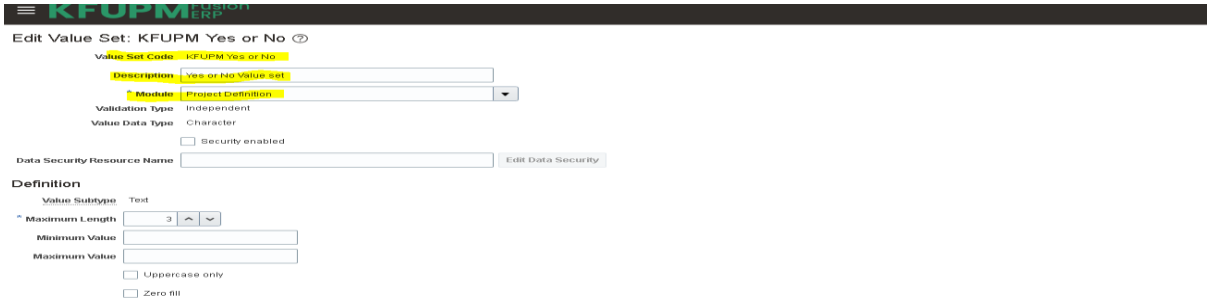
Navigation >> Setup and Maintenance >>Task >> Manage Project Task Descriptive Flexfields

Click On Pencil Icon









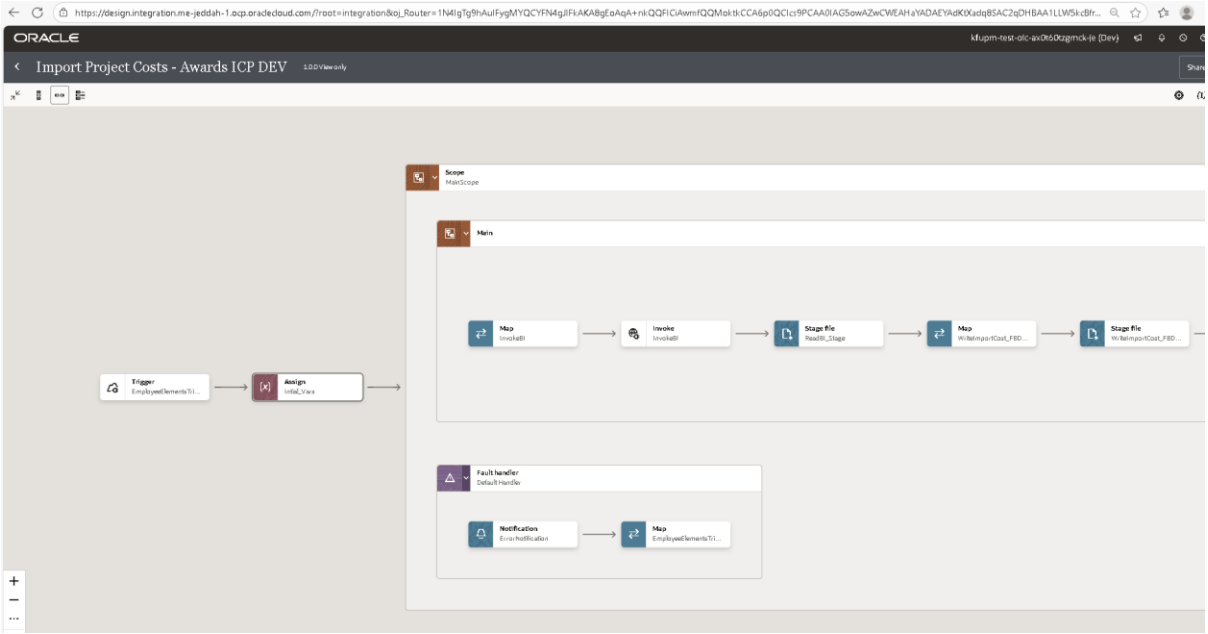
Technical Configuration :

Integration Development

- **Integration Name**: ***Import Project Costs - Awards ICP DEV***
- **Integration Type**: *App-Driven*
- **Connections Used**:
  - **REST Connection**:
    - **Purpose**: Used as a trigger to invoke the integration
    - **Connection Name**: ***KFUPM_TEST_REST_TRIG***
  - **SOAP Connection**:
    - **1.Purpose**: Used to fetch approved Project Award Compensation transactions by running the BI report from the Fusion Application
    - **Connection Name**: ***Call_BI_Dev_KFUPM_MS***
    - **2.**Purpose: To Get The Scheduler Report Info (DocumentId) And Get The Document Data
    - **Connection Name**: ***BISchedule_SOAP_KFUPM_DEV***
  - **ERP Connection**:
    - **Purpose**: Used to load the FBDI ZIP file to UCM and submit the Import Cost ESS job
    - **Connection Name**: ***DEV_KFUPM_ERP***

Variables Used in the Integration.

The image below illustrates the creation of variables and the scope defined for error handling. It also shows the invocation of the BI report, followed by the use of stage file operations to read the BI report data and write it in CSV format.

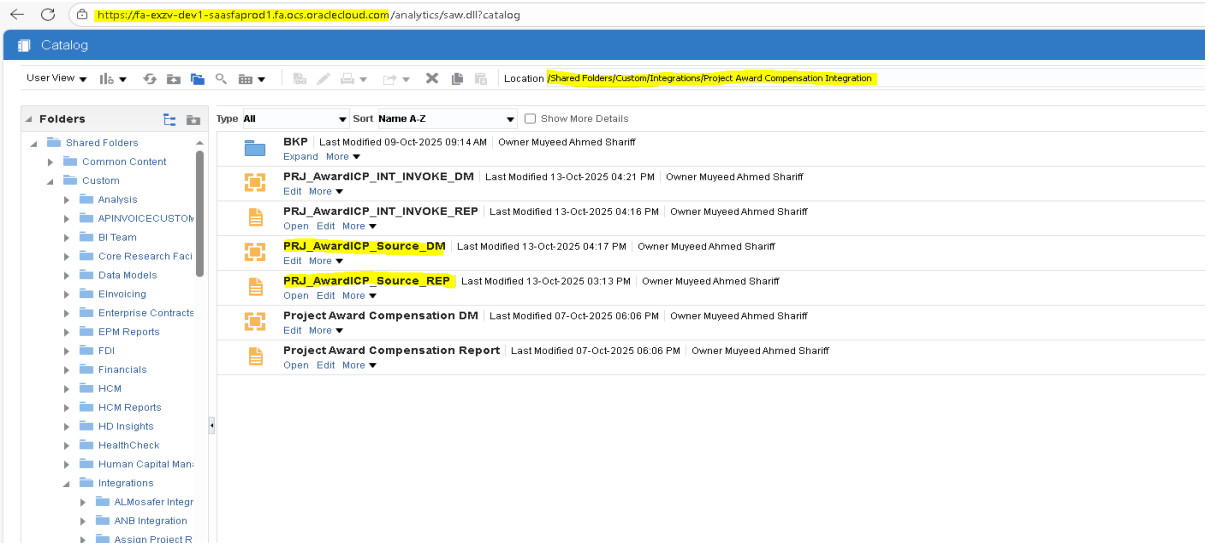


BIReportPath: **/Custom/Integrations/Project Award Compensation Integration/PRJ_AwardICP_Source_REP.xdo**

DataModelName: **PRJ_AwardICP_Source_DM**

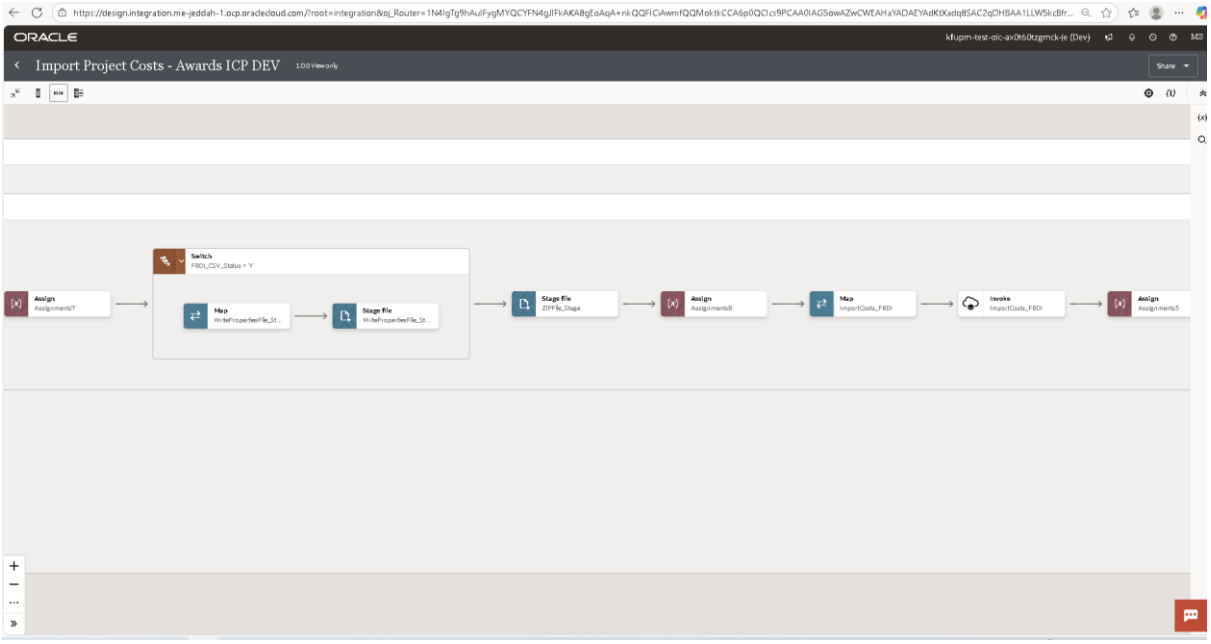
ReportName: **PRJ_AwardICP_Source_REP**

ParameterName: **p_approved_date.**



The image below illustrates the creation of a properties file used for zipping the FBDI file. It checks the status of the previous BI response operation before proceeding to zip the files. Subsequently,

the **"Import Bulk Data into Oracle ERP Cloud"** operation is used to load the zipped file into UCM, and the ESS job process ID is assigned to a variable.



The image below illustrates the use of a while loop to monitor the ESS job process status, ensuring its completion before proceeding to submit the **Import Cost ESS Job**. After submission, another while loop is used to check the status of the Import Cost ESS Job.

Parameters Used During ESS Job Submission.

JOB PACKAGE NAME: oracle/apps/ess/projects/costing/transactions/onestop

JOB DEFINITION NAME: ImportProcessParallelEssJob

PARAMETER_1 (Business Unit): KFUPM BU

PARAMETER_2 (Business Unit ID): 300000003347133

PARAMETER_3(Process Mode): IMPORT_AND_PROCESS

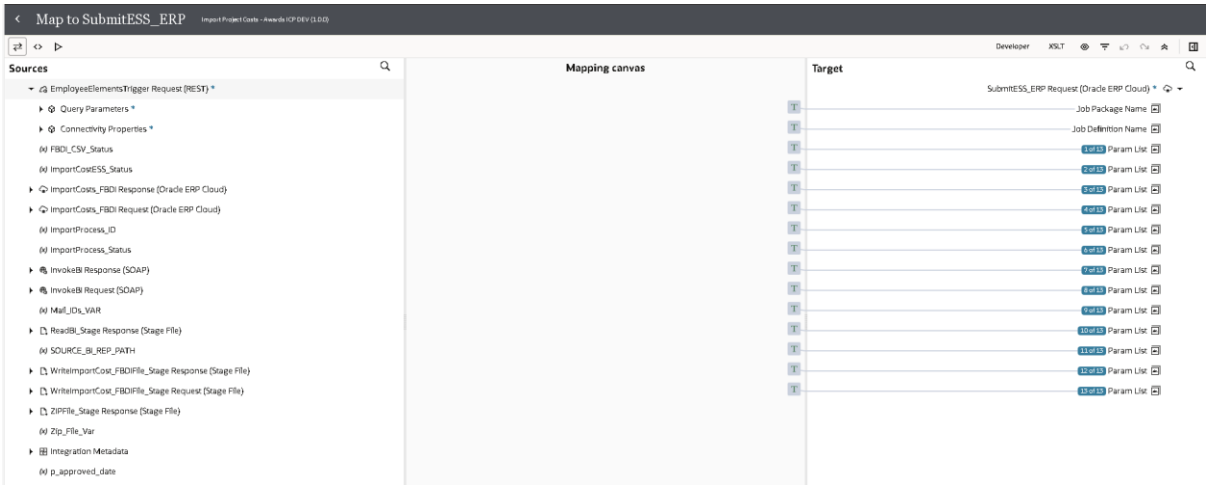
PARAMETER_4 (Transaction Status): PREV_NOT_IMPORTED

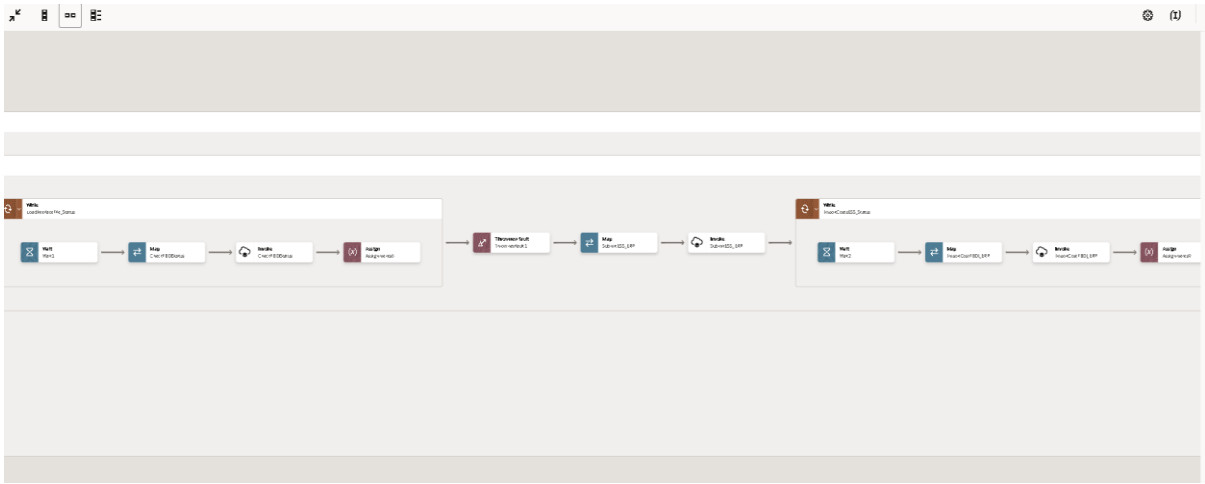
PARAMETER_6(Transaction Source): 300000879141482

PARAMETER_7(Document): 300000930104449

PARAMETER_13(Generate Report): ORA_PJC_DETAIL

PARAMETERS: - 5, 8, 9, 10, 11, 12 = #NULL

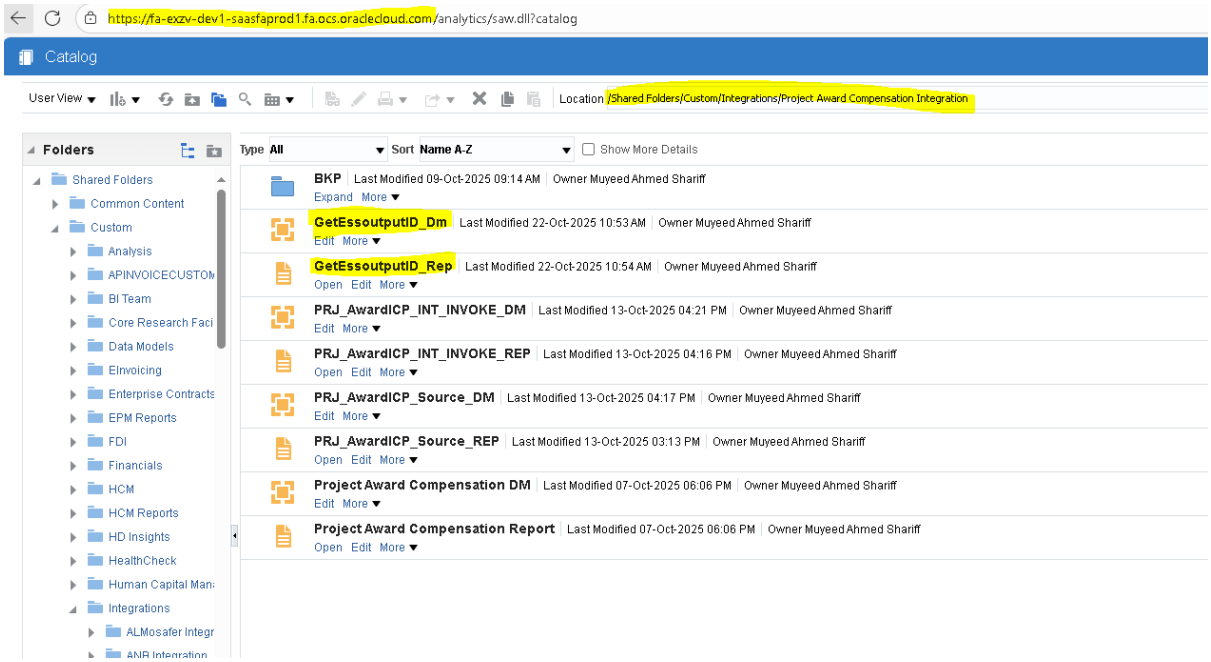


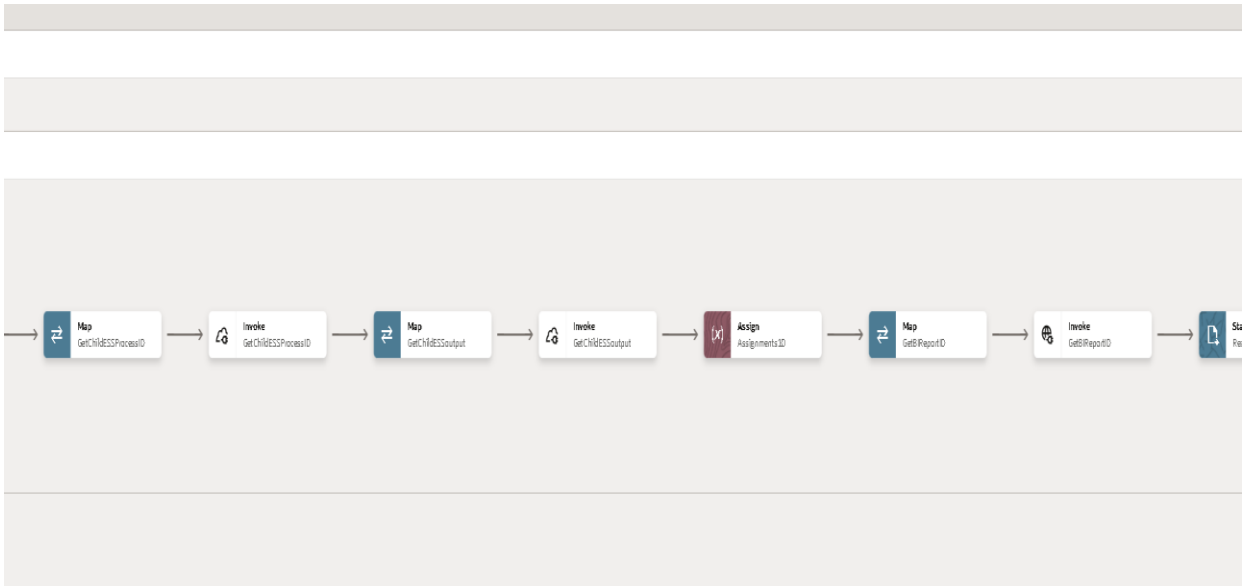


In Case Of Any Error Occurred in both the Load File Import Process and Import Costs ESS then it will throw an Error mail notification

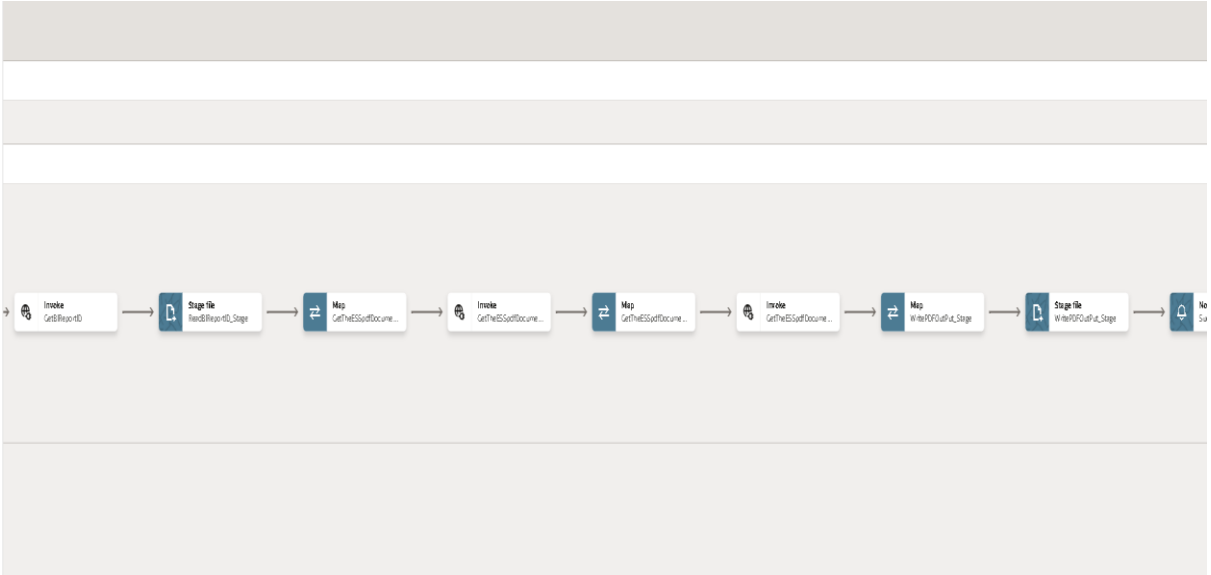
The image below illustrates that after the completion of the Import Cost ESS Job, we fetch the Flow ID, which helps us retrieve the Import Cost Child (Report Output) ESS Job Process ID. Using the Flow ID, we fetch the Child Process ID, and then use that Child Process ID to retrieve the ExternalProcessID via a custom BI report.

Custom Report: /Custom/Integrations/Project Award Compensation Integration/GetEssoutputID_Rep.xdo

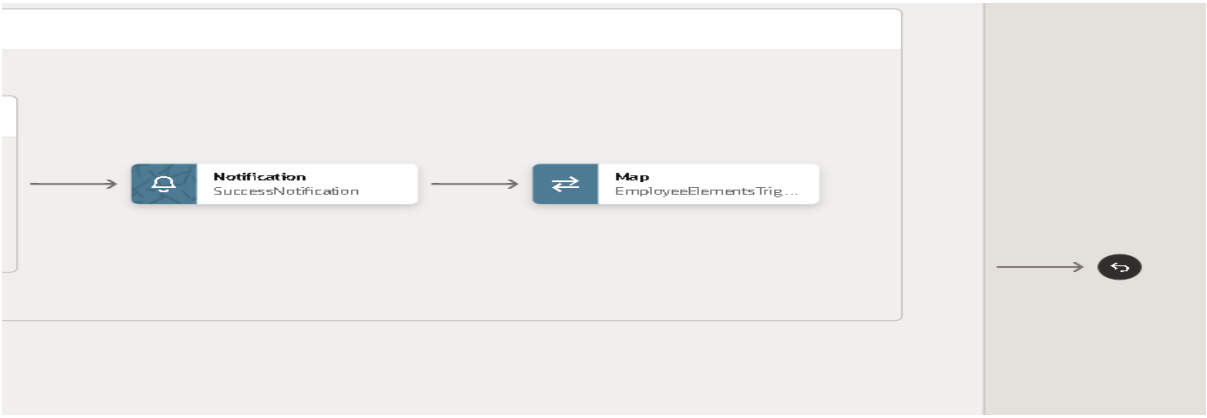




The image below illustrates that after fetching the ExternalProcessID, a SOAP Scheduler connection operation named **getScheduledReportOutputInfo** is invoked by passing the previously fetched ExternalProcessID, which returns the Document ID of the Import Costs ESS output. Using the same SOAP connection, another operation named **getDocumentData** is performed by passing the retrieved Document ID. Finally, the PDF file is written using the opaque schema of the Stage File action method 'Write File'.



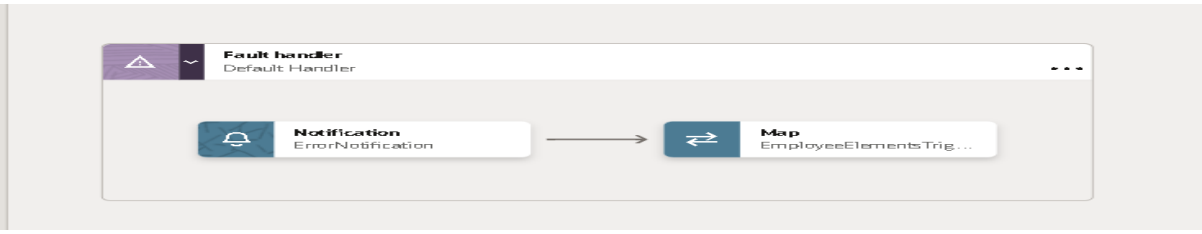
The image below illustrates that after the completion of the Import Cost ESS Job, a notification is triggered with BI Report Data (INPUT) and PDF ESS Report (OUTPUT), and the integration flow is concluded at the endpoint with a success message in XML format.



FAULT HANDLER :

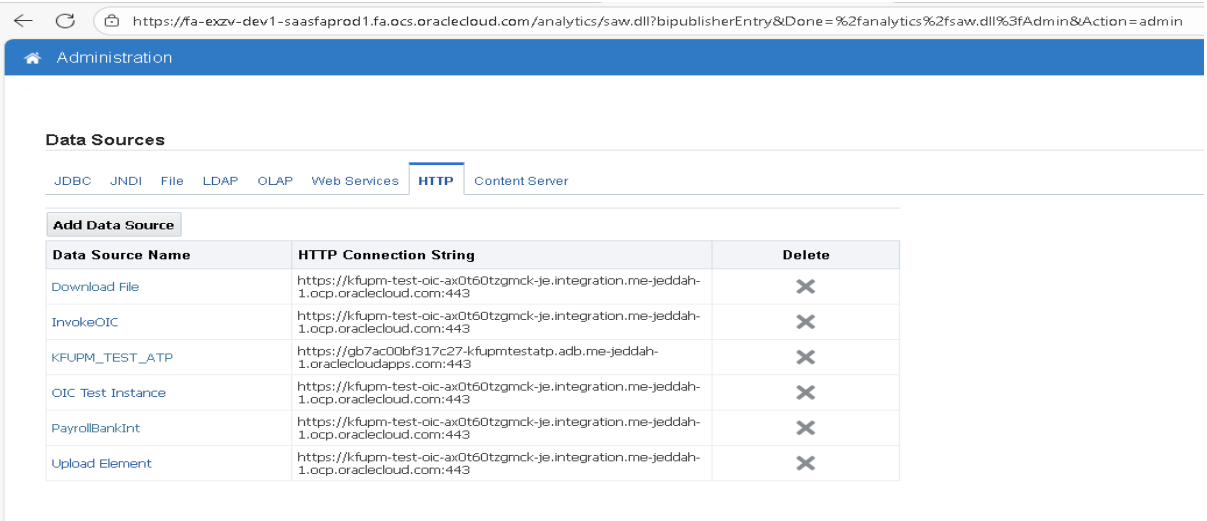
The image below illustrates the fault handler. If any error occurs in the above integration flow, an email notification is triggered, and the integration flow is concluded at the endpoint with a response

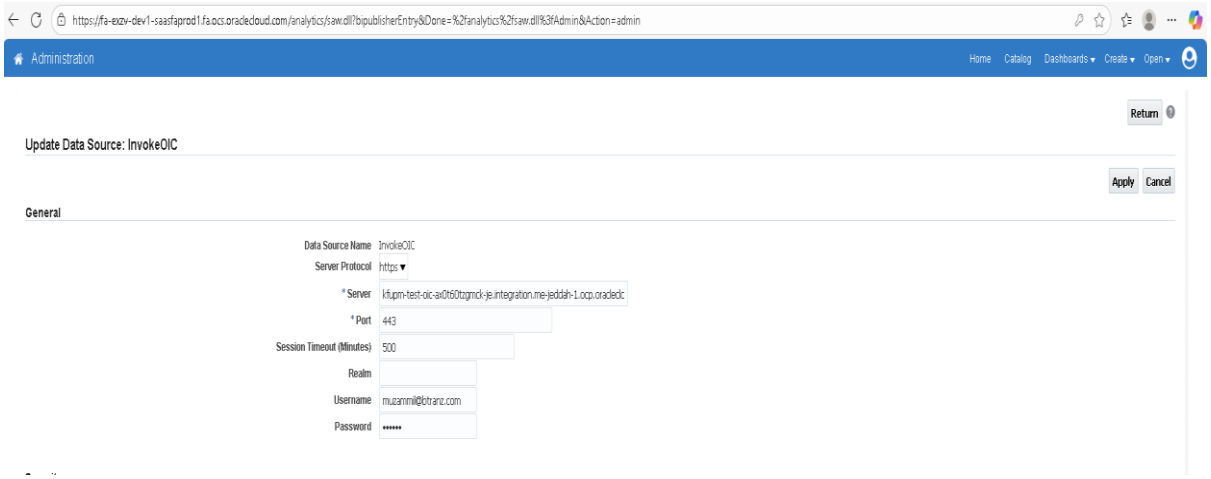
marked as **ERROR**, including the error details in XML format.



Custom ESS Job Configuration

Create an HTTP Data Source in BI Administration of the OIC Instance > Manage Publishers>Data Sources>Add Data Source

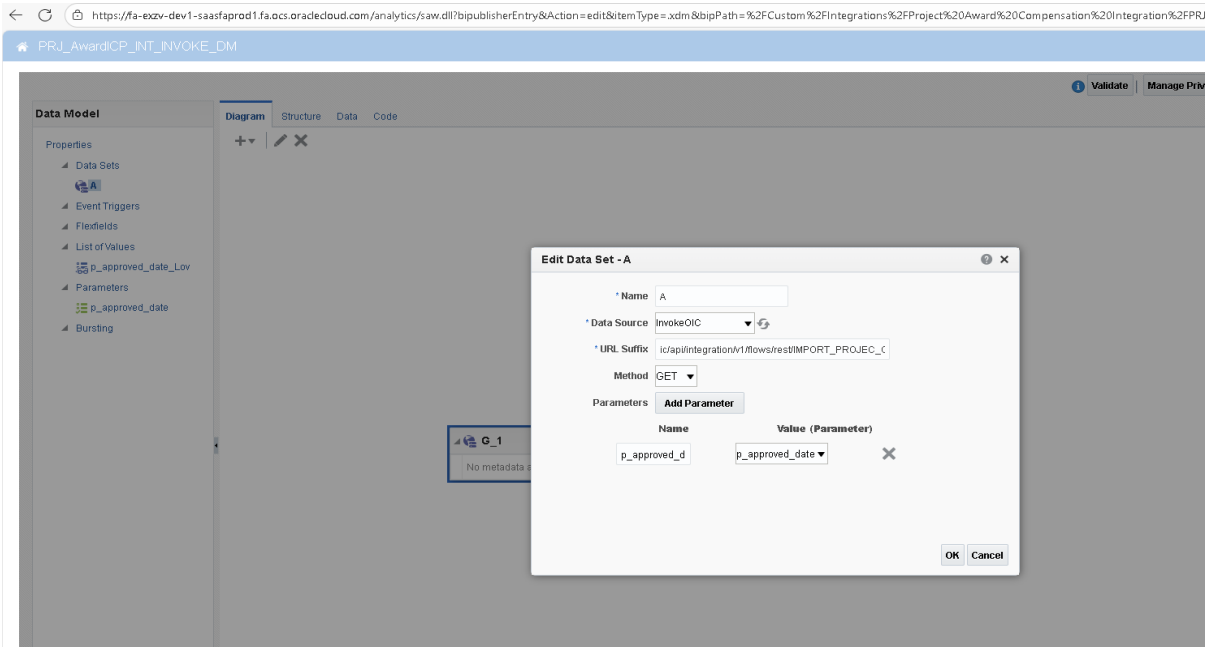




Server : **kfupm-test-oic-ax0t60tzgmck-je.integration.me-jeddah-1.ocp.oraclecloud.com**

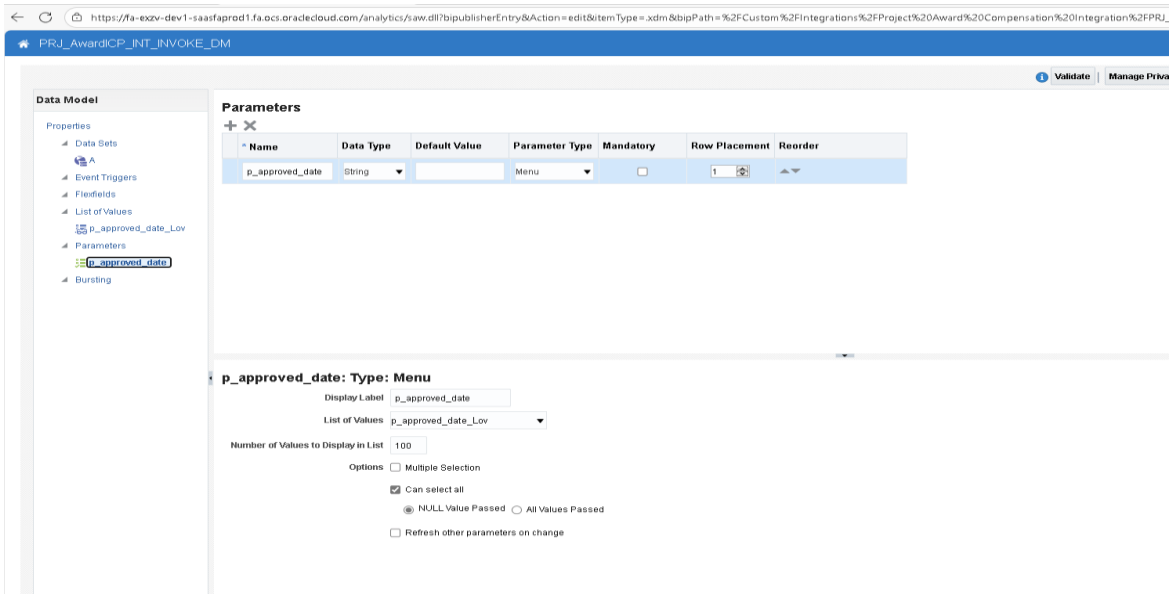
Create a Data Model for the HTTP (XML Feed) Dataset

Utilize the HTTP data source previously created in BI Administration to build a data model for the HTTP (XML Feed) dataset.



URL Suffix: It's The Integration End Point.

ic/api/integration/v1/flows/rest/IMPORT_PROJEC_COSTS_AWARDS_ICP/1.0/ImportProjectCosts



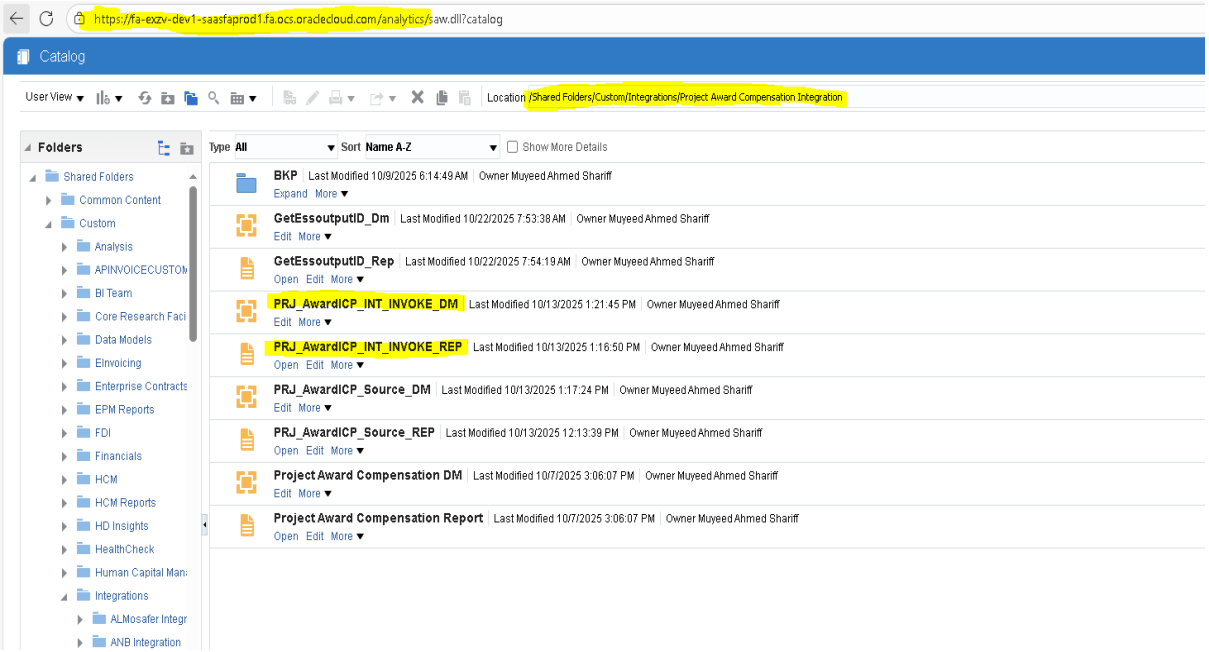
Note:

Ensure that the parameter name (*p_approved_date*) is same across all related objects.

Report Parameter LOV Query:

```
WITH months AS (
  SELECT ADD_MONTHS(DATE '2020-01-01', LEVEL - 1) AS month_date
  FROM dual
  CONNECT BY ADD_MONTHS(DATE '2020-01-01', LEVEL - 1) <= TRUNC(SYSDATE, 'MM')
)
SELECT TO_CHAR(month_date, 'MON-YYYY', 'NLS_DATE_LANGUAGE=ENGLISH') AS month_year
FROM months
ORDER BY month_date
```

Now, create the report using this data model, which can be used for creating the custom ESS job.



Report path : - /Custom/Integrations/Project Award Compensation Integration/PRJ_AwardICP_INT_INVOKE_REP.xdo

<u>Note :</u>

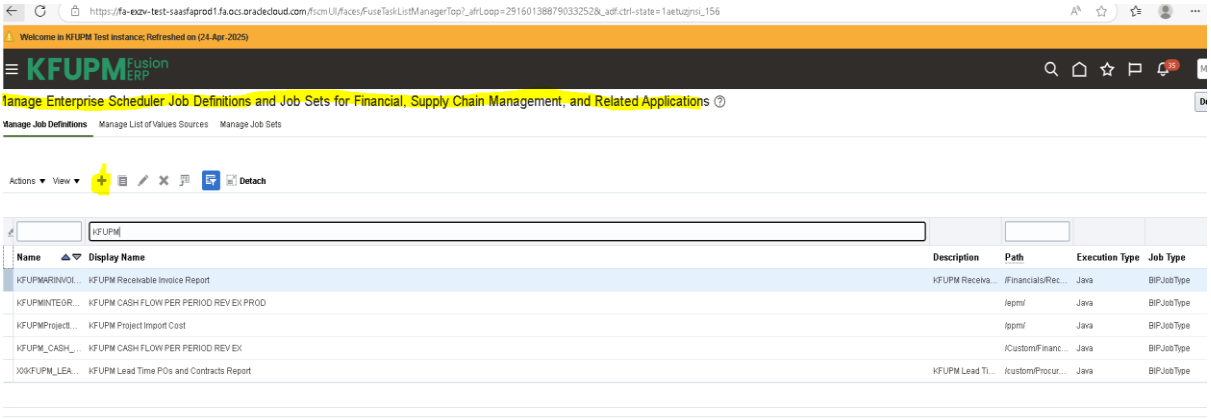
This report triggers the created integration and can be executed using a custom ESS job.

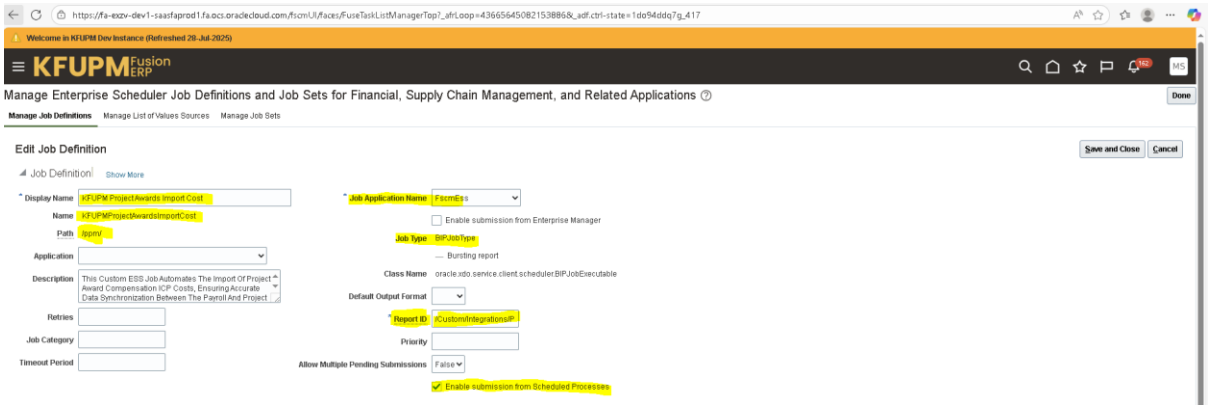
Now, Navigate To Setup and maintenance > Tasks> Search for Task.

Task Name: Manage Enterprise Scheduler Job Definitions and Job Sets for Financial, Supply Chain Management, and Related Applications

Click Add (+) Icon

PPM To Payroll Integration
(Project Award Compensation)





JobApplicationName: FscmEss

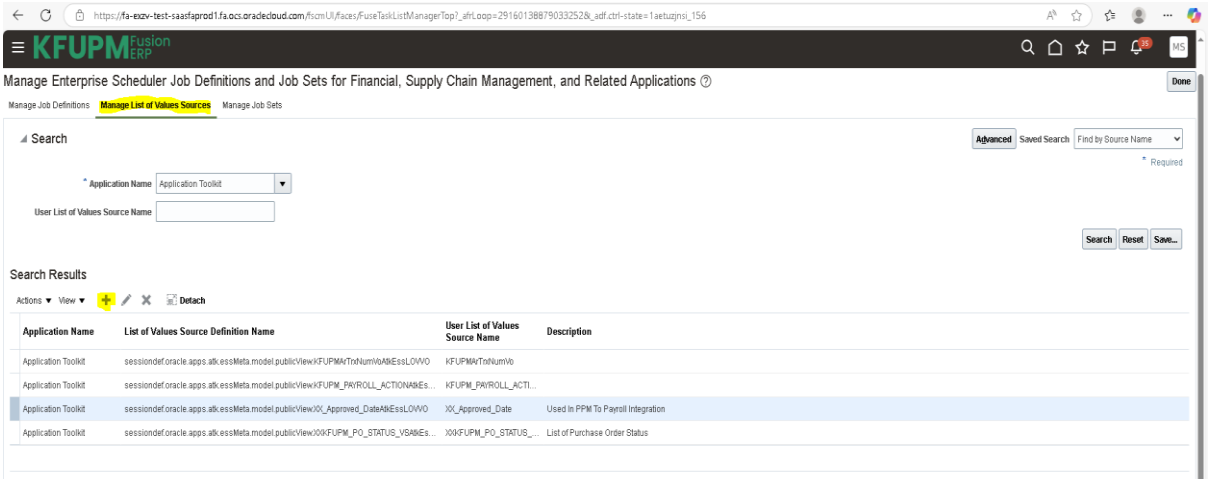
Job Type: BIPJobType

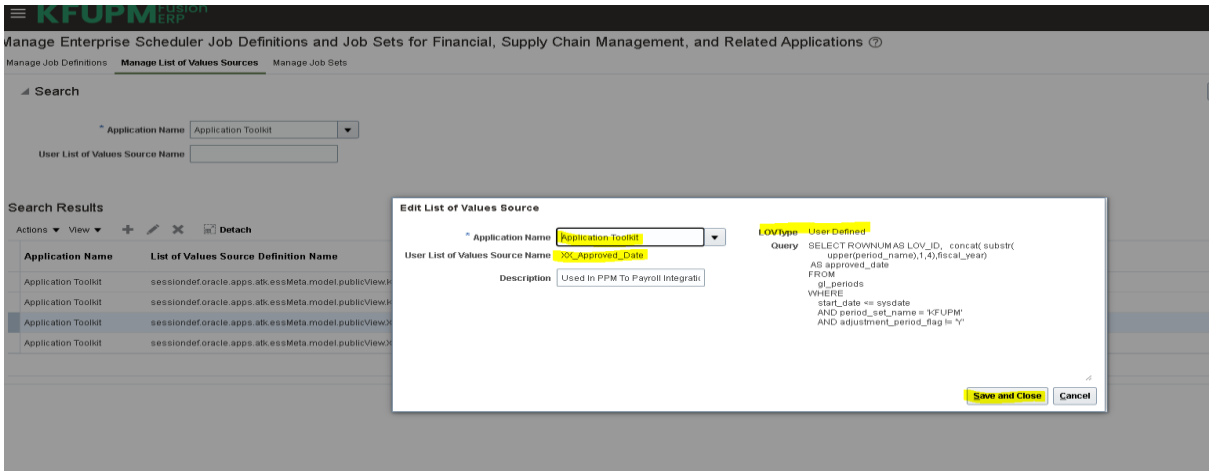
Path:/ppm/

Report ID :

/Custom/Integrations/Project Award Compensation Integration/PRJ_AwardICP_INT_INVOKE_REP.xdo

Create the **List of Values (LOV)** source to be used in parameter definitions for the custom ESS job.

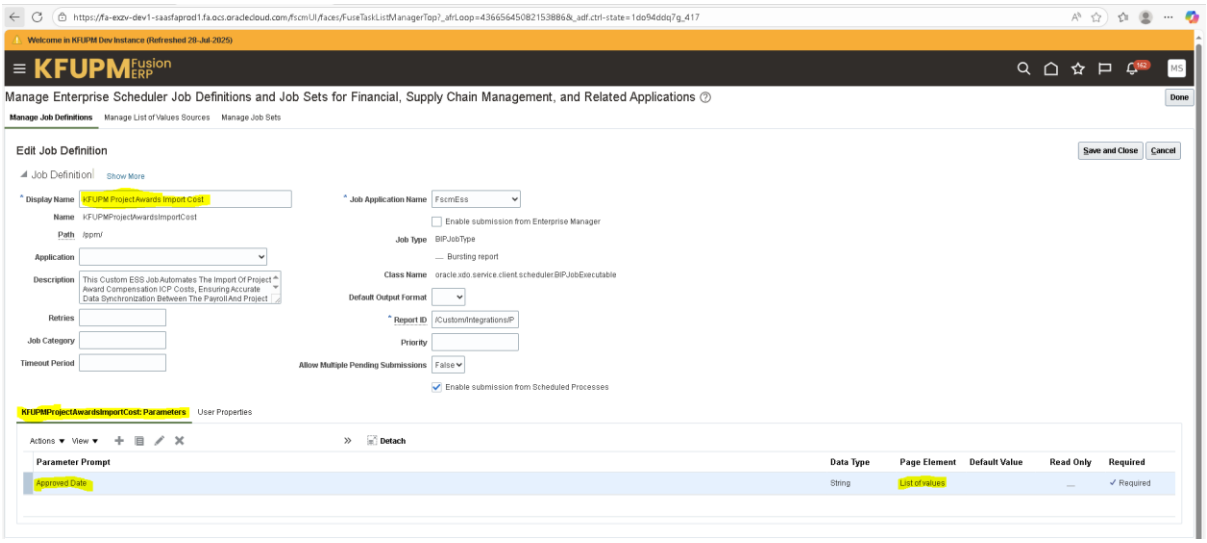


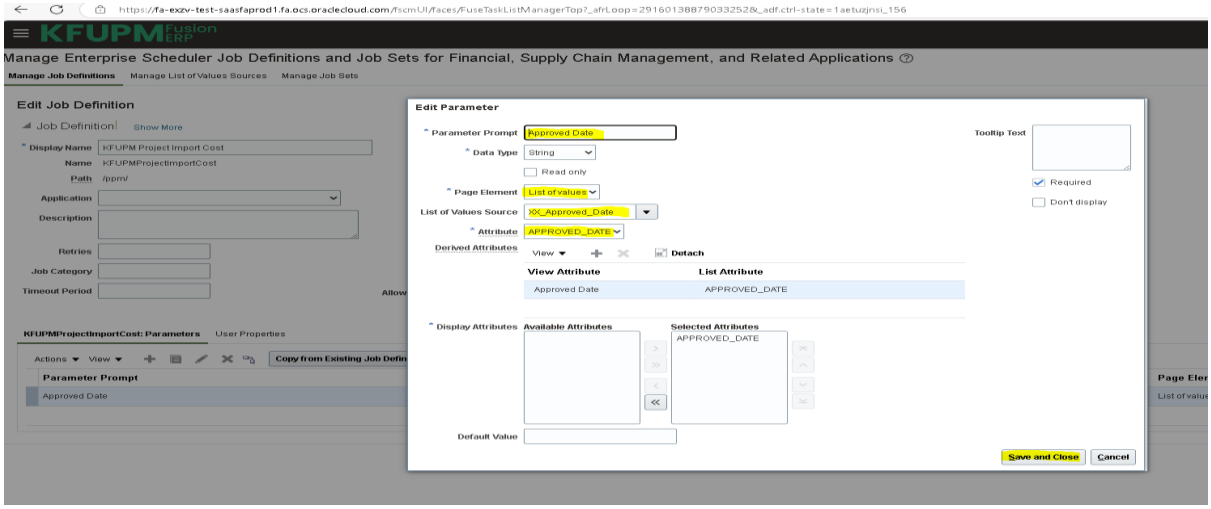


ESS LOV Query:

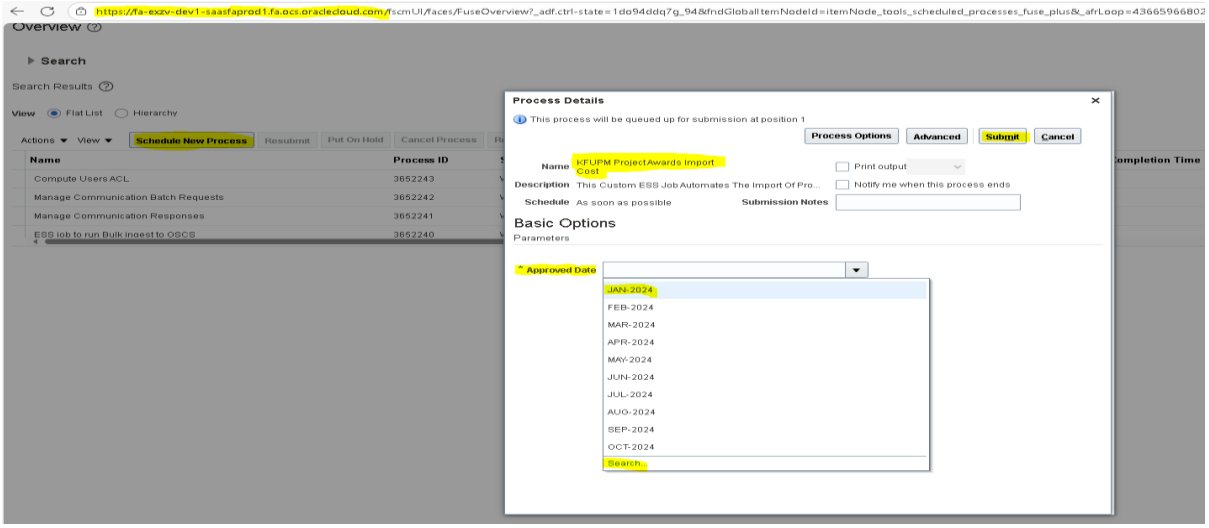
```
SELECT ROWNUM AS LOV_ID,   concat( substr(
       upper(period_name),1,4),fiscal_year)
 AS approved_date
FROM
   gl_periods
WHERE
   start_date <= sysdate
   AND period_set_name = 'KFUPM'
   AND adjustment_period_flag != 'Y'
```

Now, use this LOV source while configuring the ESS job parameter.





After successfully creating the custom ESS job, it will appear in the Scheduled Processes, and you will be able to pass the Approved Date parameter while submitting the ESS job, as shown below.



Frequently Raised Error

Error: **You must provide a value for at least one of the following attributes: Task ID, Task Name, and Task Number.**

Reasons: **The ICP Indicator (DFF) Is NULL.**

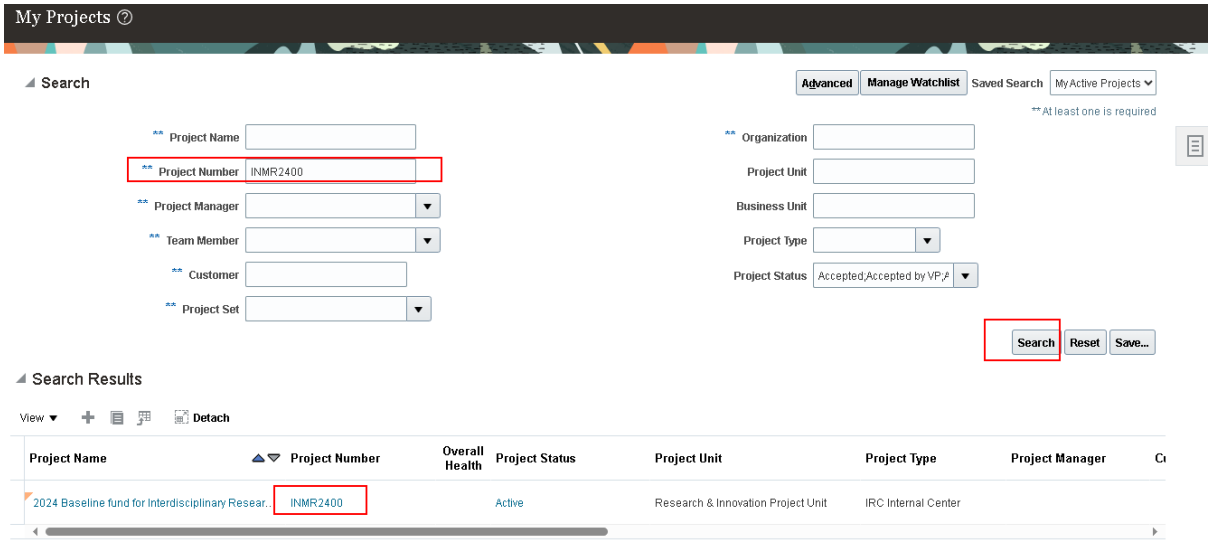
The ICP Indicator (DFF) Is Set To 'NO'.

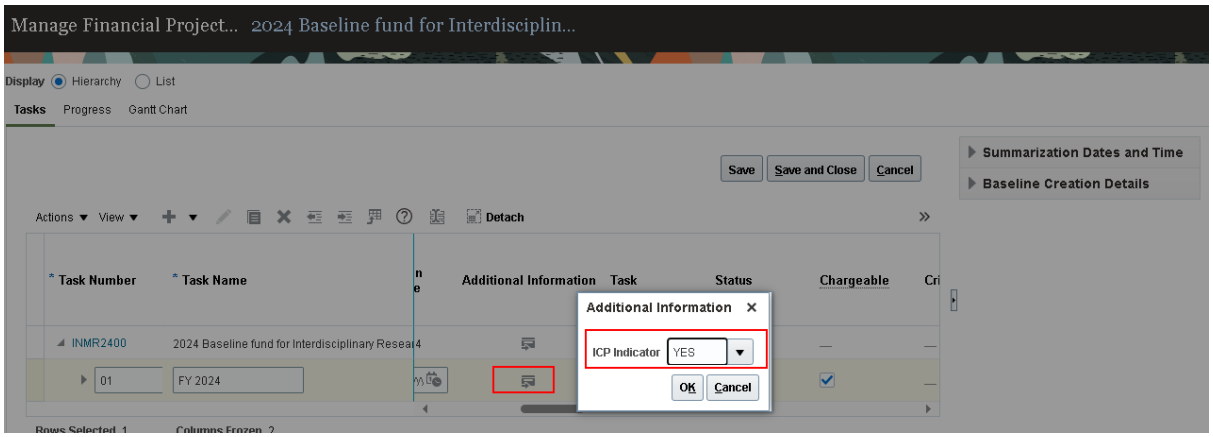
Multiple Tasks having ICP Indicator Set To 'YES'.

Note: **The Task Number will be fetched only if exactly one task has the ICP Indicator set to 'YES'.**

Solution: **Set ICP Indicator To Respective Task**

Navigation ---> Projects ---> Project Financial Management ---->Click on Task ----> Manage Projects --> Enter Project Number ---> Click on Search --->Task --> Manage Financial Project Plan.





SQL Query :

PRJ_AwardICP_FeedingQuery.sql

• The query fetches all **Approved Project Award Compensation transactions** for a given month, based on the input parameter in the format **MON-YYYY**. (i.e; MAY-2025)

• It specifically filters for transactions **not yet imported** into the **PPM module**, where the **Plan Name** is **Project Award Compensation**.

• The logic handles scenarios where a **project has multiple tasks** with the **ICP Indicator set to 'YES'**.

- In such cases, the transaction will still be fetched, but the **Task Number will be returned as NULL**.

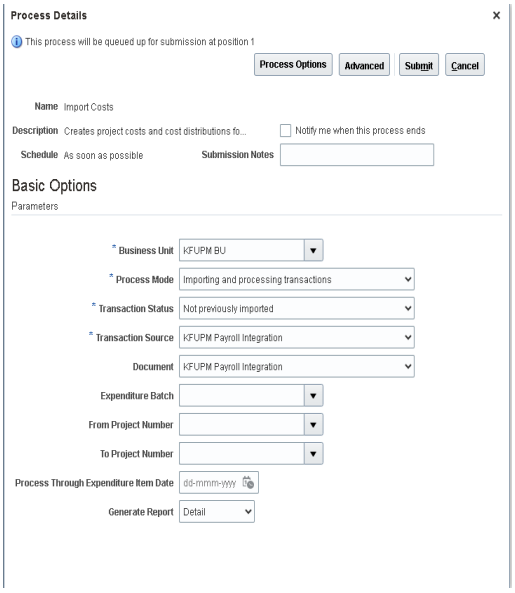
• The query includes logic to check for **already imported transactions** by referencing the **original transaction reference**.

• Note**:** Every column mapped to the **FBDI file** is sourced **only from the BI report**.

- Any future modifications should be made to the **BI report**, which will automatically reflect in the integration.
- No feeding data has been hardcoded in the integration.

• **Note:** If any **parameter values for the standard Import Cost ESS Job** (as shown in the image below) need to be modified:

- These changes must be made in the **integration** at the node, where the **ESS Job is submitted**.
- This Modification has been documented as below.



PARAMETER_1 (Business Unit): KFUPM BU

PARAMETER_2 (Business Unit ID): 300000003347133

PARAMETER_3 (Process Mode): IMPORT_AND_PROCESS

PARAMETER_4 (Transaction Status): PREV_NOT_IMPORTED

PARAMETER_6 (Transaction Source): 300000879141482

PARAMETER_7 (Document): 300000930104449

PARAMETER_13 (Generate Report): ORA_PJC_DETAIL

PARAMETERS: - 5, 8, 9, 10, 11, 12 = #NULL